Lecture 6:
Phase Retrieval



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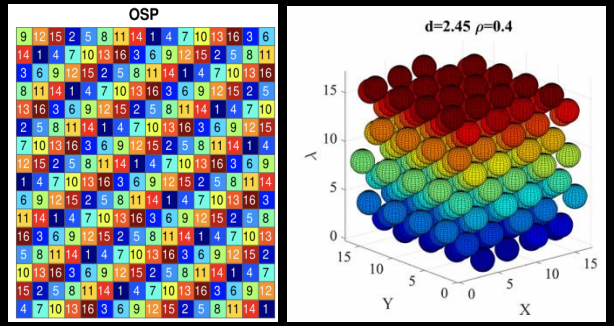
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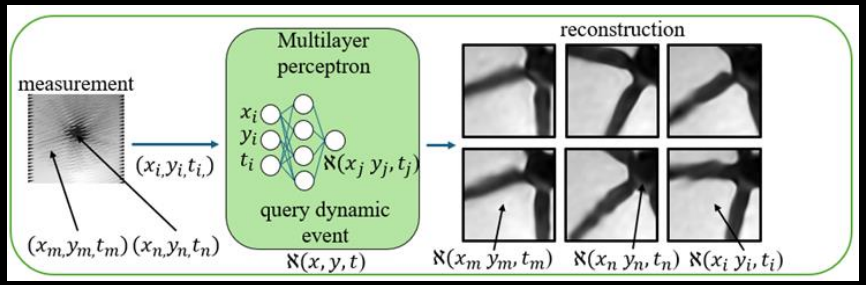
Computational Photography: Applications

- It can be applied to different signals
- It use compressive measurements
- To solve an inverse problem

Multispectral



Video

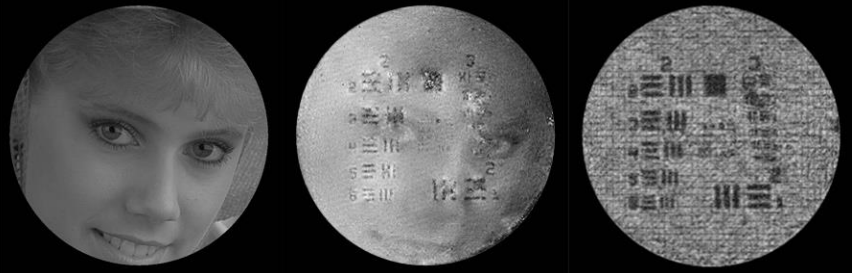


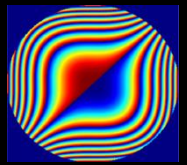
GT phase

Recovered phase

Recovered Amplitude

A woman





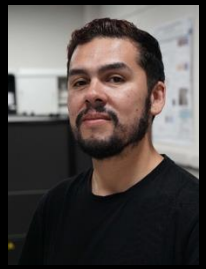
1995

Cubic Phase



Depth

Phase Retrieval by Designed Hadamard Complementary Coded Apertures










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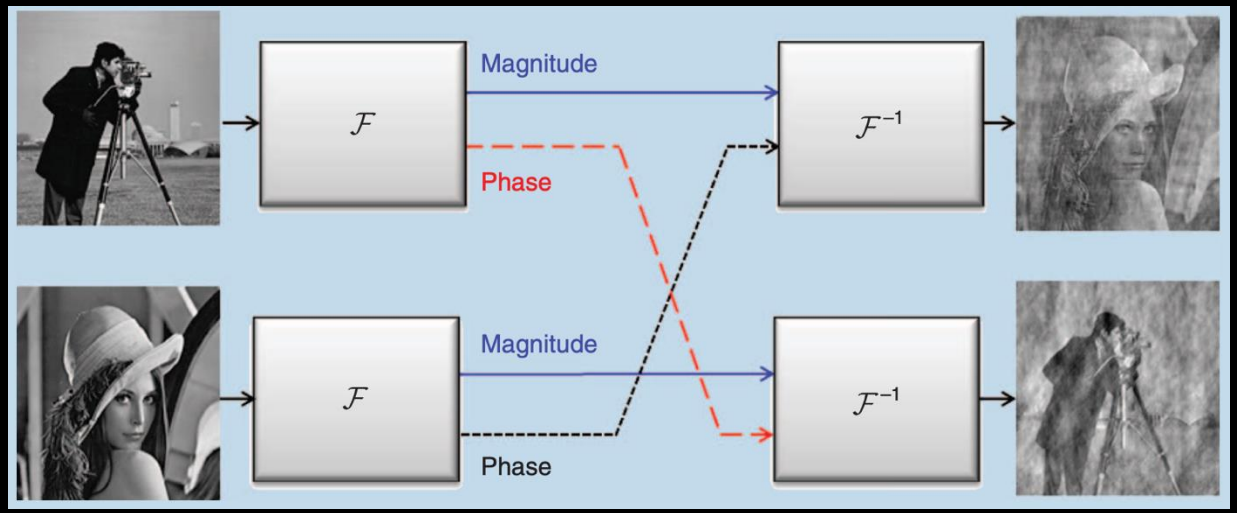
OPTOLAB

Phase Retrieval

Phase retrieval is the problem of reconstruction of a signal or object from the magnitude of its Fourier transform, where the phase information is missing.

Why is the phase important?

The magnitude alone does not uniquely determine the signal; phase encodes crucial structural information.



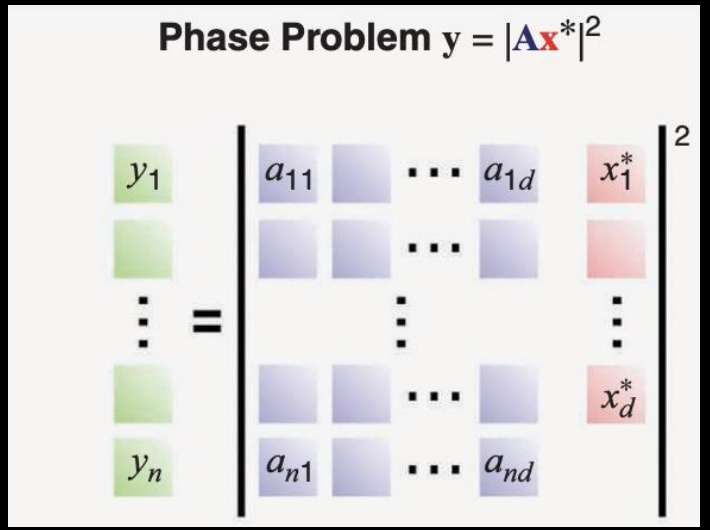
The result clearly demonstrates the importance of phase information for image recovery.

Shechtman, Y., Eldar, Y. C., Cohen, O., Chapman, H. N., Miao, J., & Segev, M. Phase retrieval with application to optical imaging: a contemporary overview. *IEEE signal processing magazine*, *32*(3), 87-109, 2015.

Phase Retrieval

- Let $x^* \in \mathbb{C}^n$ be the unkown signal.
- Given $y_i = |\langle a_i, x^* \rangle|^2$ for $i = 1, \dots, m$

- Goal: Recover x^* (up to a global phase ambiguity)
- The problem is non-convex, making it computationally challenging
- Phase retrieval has significant applications

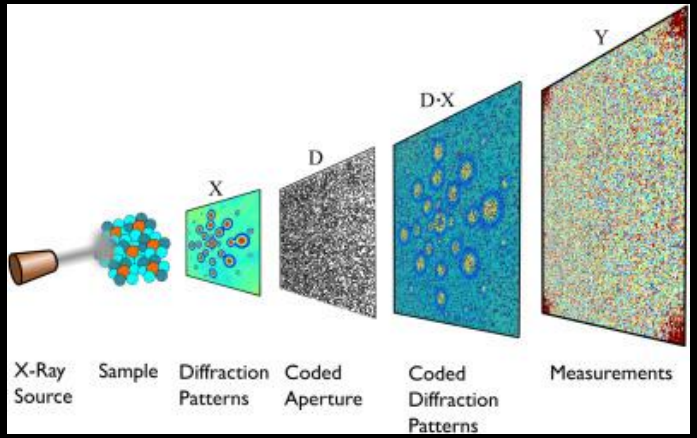


Dong, J., Valzania, L., Maillard, A., Pham, T. A., Gigan, S., & Unser, M. Phase retrieval: From computational imaging to machine learning: A tutorial. *IEEE Signal Processing Magazine*, *40*(1), 45-57, 2023.

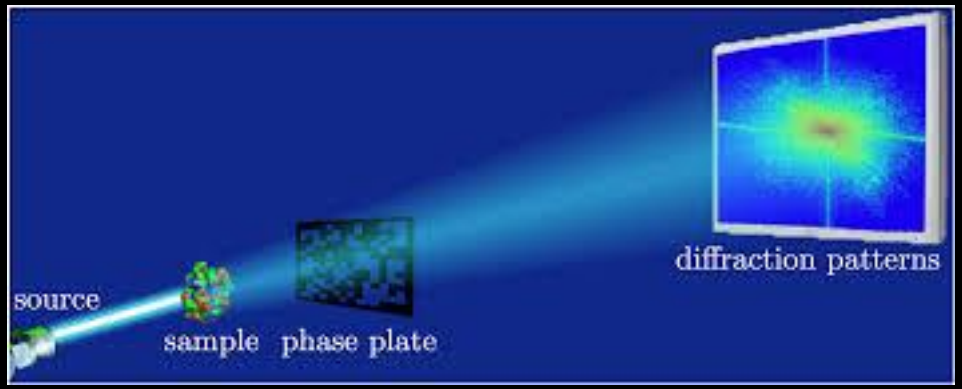
Phase Retrieval Applications

Where this happens

Many physical systems (optics, X-ray Crystallography, astronomy) can only measure intensity (i.e., squared magnitude) not phase.



X-ray-Crystallography



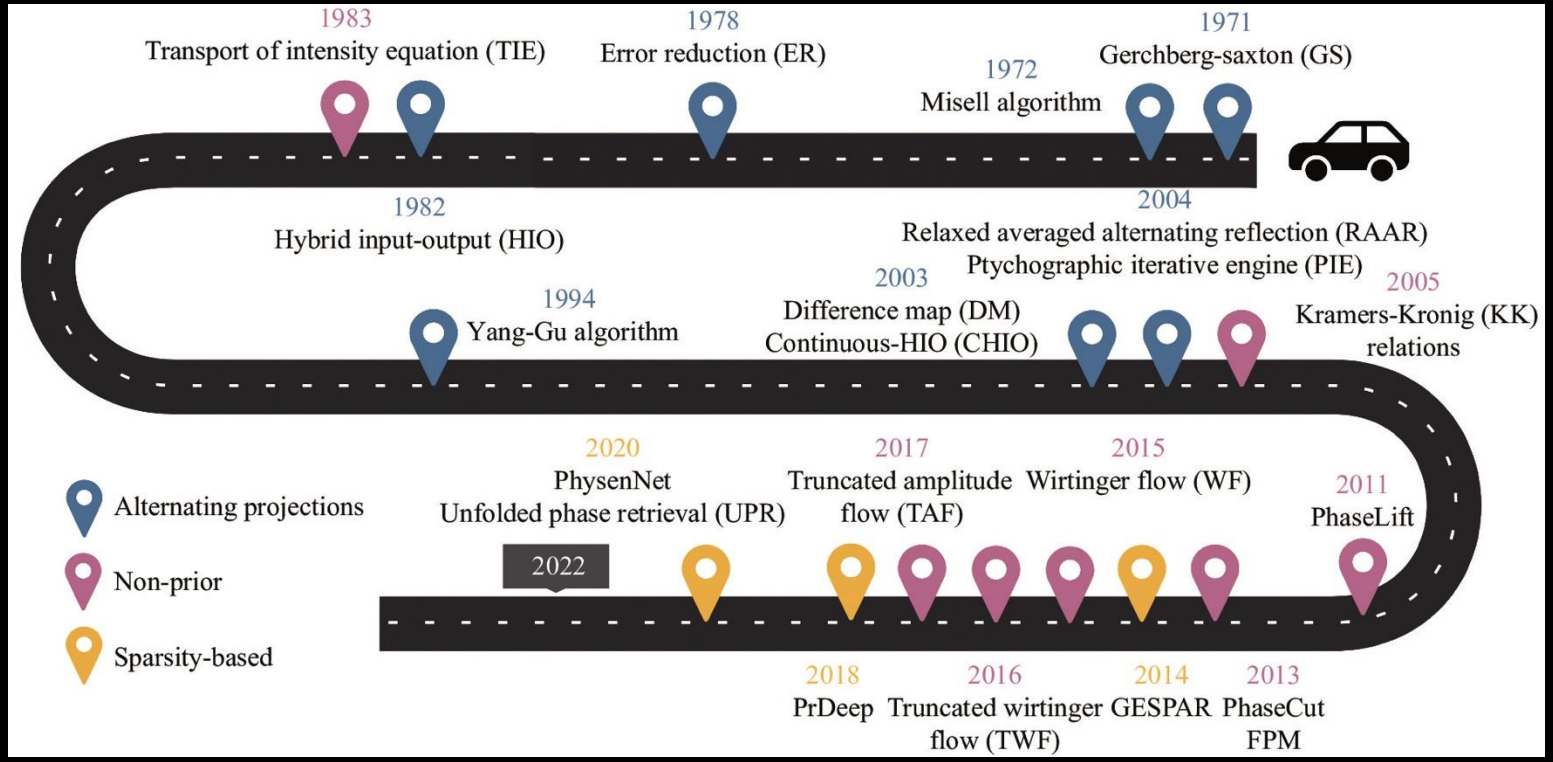
X-ray-Crystallography

Reconstruction algorithms

Pinilla, S., García, H., Díaz, L., Poveda, J., & Arguello, H. Coded aperture design for solving the phase retrieval problem in X-ray crystallography. *Journal of Computational and Applied Mathematics*, *338*, 111-128, 2018.

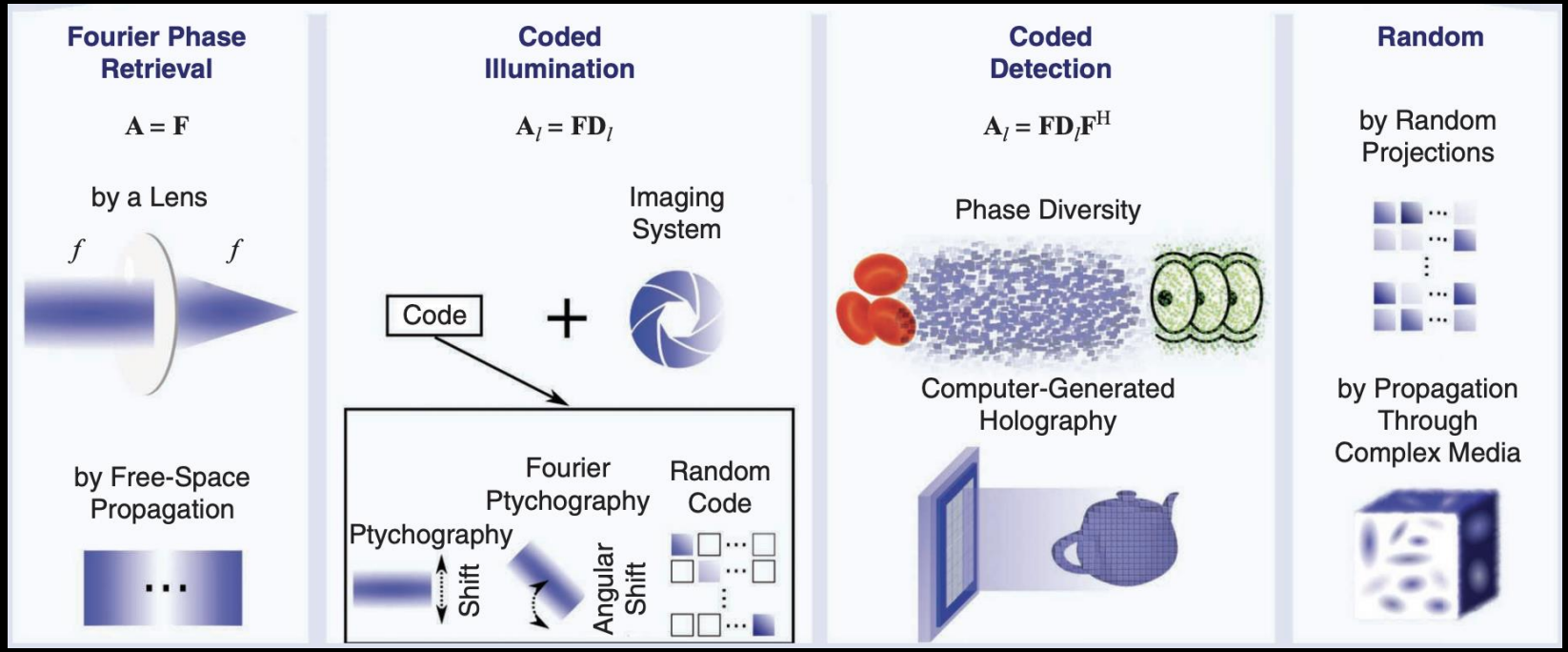
Candes, E. J., Li, X., & Soltanolkotabi. Phase retrieval from coded diffraction patterns. *Applied and Computational Harmonic Analysis*, *39*(2), 277-299, 2015.

Timeline Phase Retrieval Algorithms



Wang Aiye, Pan An, Ma Caiwen, Yao Baoli. Phase retrieval algorithms: principles, developments and applications. *Infrared and lasers Engineering*, *51*(12), 20220402 2022.

Classes of Phase-Retrieval Problems



Dong, J., Valzania, L., Maillard, A., Pham, T. A., Gigan, S., & Unser, M. Phase retrieval: From computational imaging to machine learning: A tutorial. *IEEE Signal Processing Magazine*, *40*(1), 45-57, 2023.

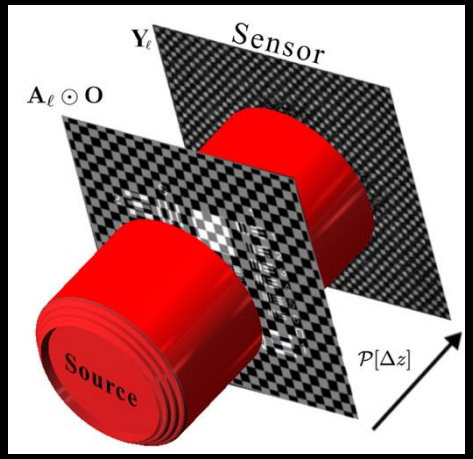
Coded Illumination Discrete Model

The ℓth coded measurement (CM) of the diffraction pattern, recorded at the image sensor is proportional to the absolute square of complex wavefront, as follows:

$$\mathbf{Y}_\ell = |\mathrm{I}_\ell|^2 + \Omega,$$

I_ℓ is the diffraction pattern
Ω is an additive noise term

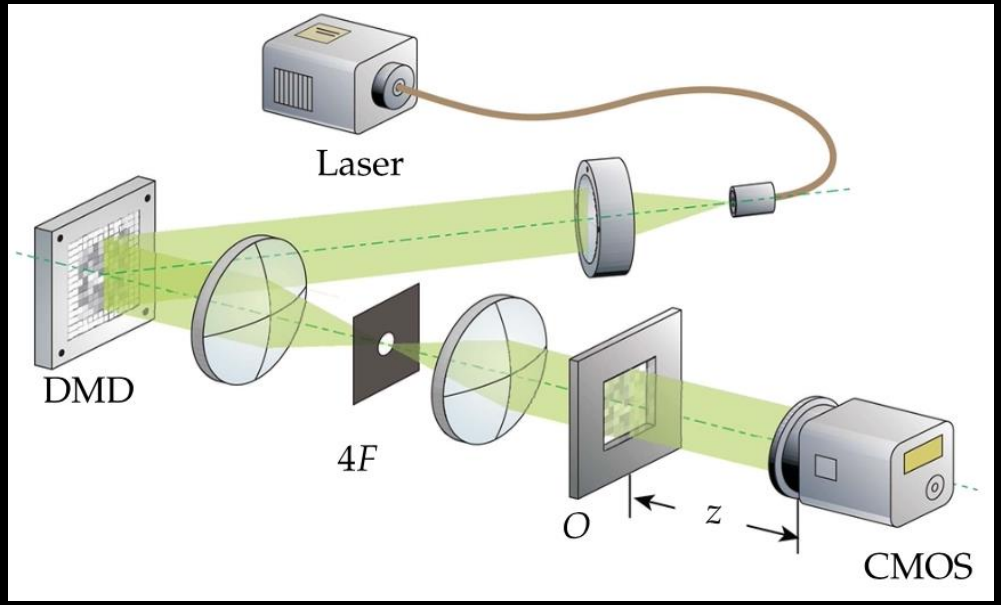
$$I_\ell = \wp[\Delta z]\{A_\ell \odot \mathbf{O}\},$$

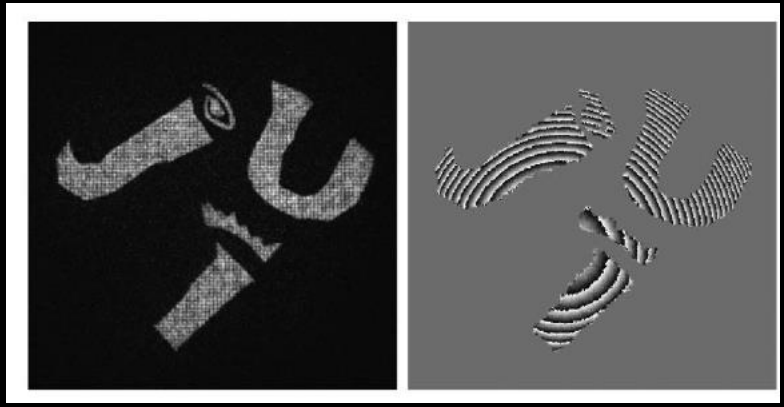


- $\wp[\Delta z]$ is the operator that represents the near-field propagator.
- A_ℓ is a set of coded apertures (CA), $A_\ell \in \{\mathbf{0}, \mathbf{1}\}^{\boldsymbol{d}\times\boldsymbol{d}}$, with $\boldsymbol{d}\times\boldsymbol{d}$ denoting the number of pixels in the CA.
- A collimated coherent beam illuminates an object, producing an unknown complex wavefront $\mathbf{O} \in \mathbb{C}^{\boldsymbol{d}\times\boldsymbol{d}}$

Previous Work in Coded Illumination

The phase is recovered using up to 20 complementary random coded apertures, which restrict agile applications.





Amplitude Phase

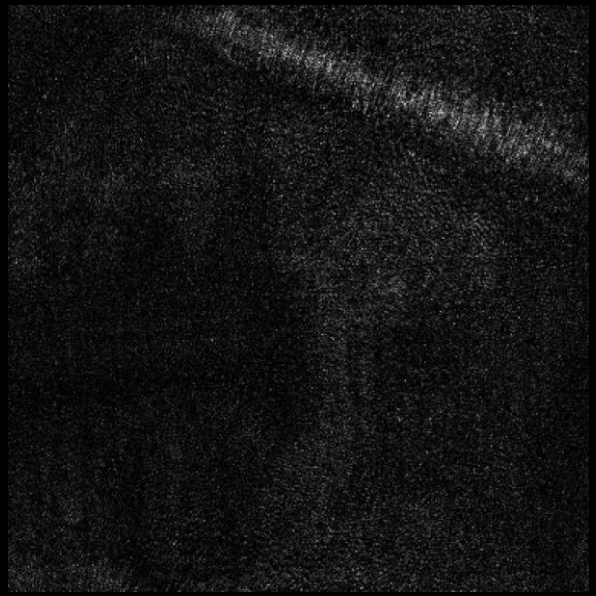
Disadvantage: Use 20 Cas, which also is demanding in the reconstrucction

Peters, E., Clemente, P., Salvador-Balaguer, E., Tajahuerce, E., Andrés, P., Pérez, D. G., & Lancis, J. Real-time acquisition of complex optical fields by binary amplitude modulation. *Optics letters*, *42*(10), 2030-2033, 2017.

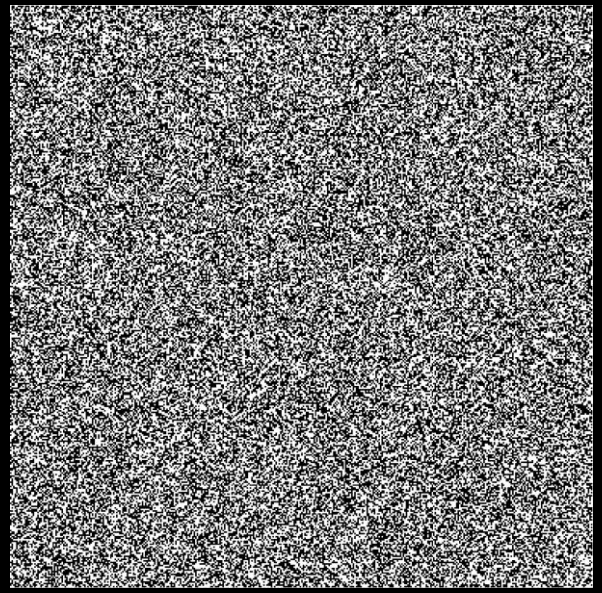
Coded illumination in the near-field

$$\mathbf{Y}_\ell = |\mathrm{I}_\ell|^2 + \Omega, \qquad I_\ell = \wp[\Delta z]\{A_\ell \odot \mathbf{O}\},$$

designing the encoding is critical for real-time applications



measurement



Traditional random coded aperture



scene

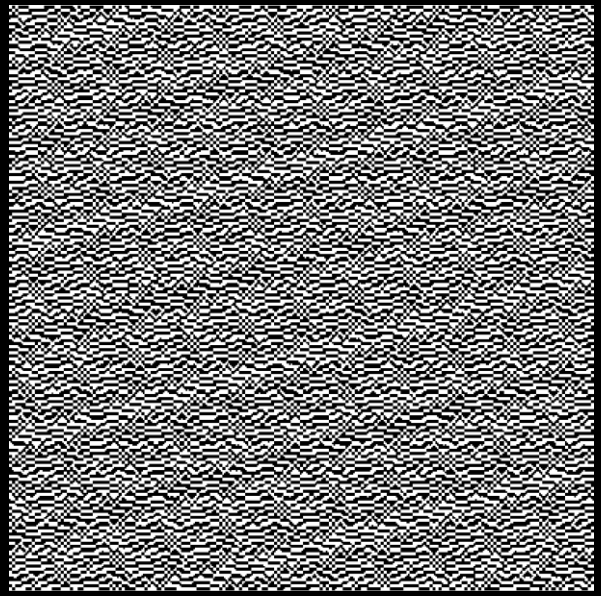
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measurement



Designed Hadamard complementary coded apertures



scene

Simulation

measurement reconstructions measurement reconstructions



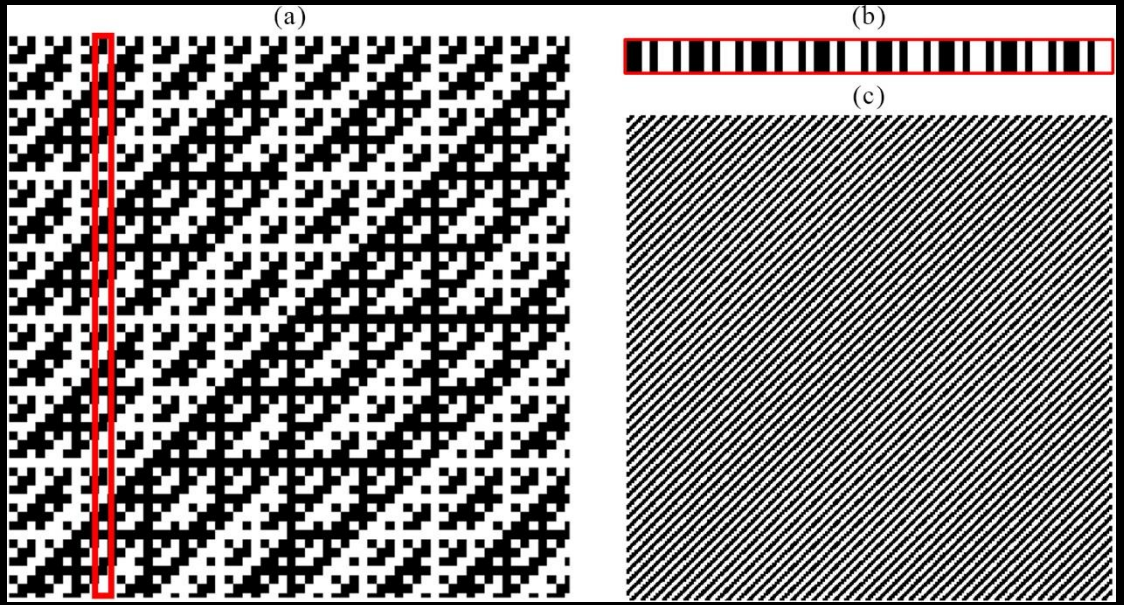
Parameters

Using 23 images of Kodak dataset

- 16 CAs 50 iterations

Hadamard Complementary Coded Apertures (HCCA)

- The proposed coded aperture set is constructed by selecting column vectors $\phi_\ell \in \mathbb{R}^{N\times 1}$ with $\ell = \{1, \dots, L\}$ from the Hadamard matrix, optimizing for the lowest condition number. Given $\mathbf{\Phi}_L = [\Phi_1, \dots, \Phi_L] \in \{0,1\}^{N\times L}$, we compute the covariance matrix $\mathbf{\Sigma} = \mathbf{\Phi}_L^T \mathbf{\Phi}_L$ and minimize its condition number, $\kappa(\mathbf{\Sigma}) = \|\mathbf{\Sigma}\|\|\mathbf{\Sigma}^{-1}\|$



Guarantees

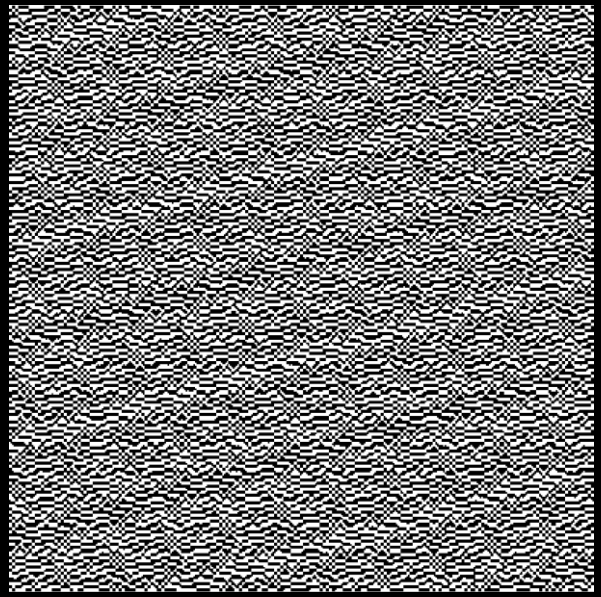
- The coded apertures exploits the a reduced condition number.
- Our design guarantees Complementarity between pairs of coded apertures

Hybrid Input-output Retrieval

$$\mathbf{Y}_\ell = |\mathrm{I}_\ell|^2 + \Omega, \qquad I_\ell = \wp[\Delta z]\{A_\ell \odot \mathbf{O}\},$$



Intensity



Designed Hadamard complementary coded apertures



Phase

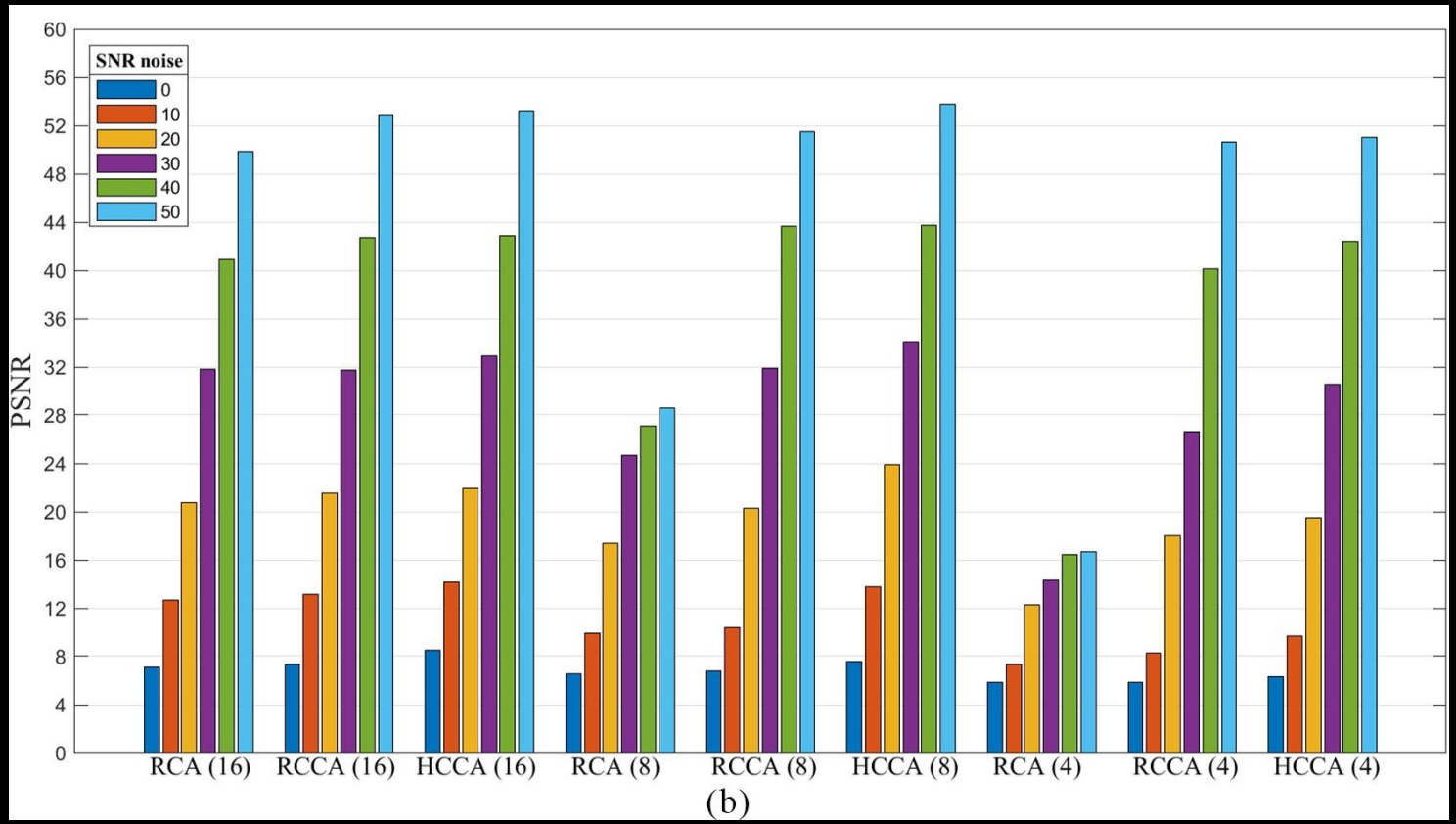
Simulation: Amplitude Quality (PSNR)

Parameters

Using 23 images of Kodak dataset
Algorithm: HIO

- 16 CAs 50 iterations
- 8 CAs 150 iterations
- 4 CAs 300 iterations

	Type of coded aperture		
Number of Masks	RCA	RCCA	HCCA
16	9.1041	9.0965	9.0537
8	5.0229	5.0295	4.9304



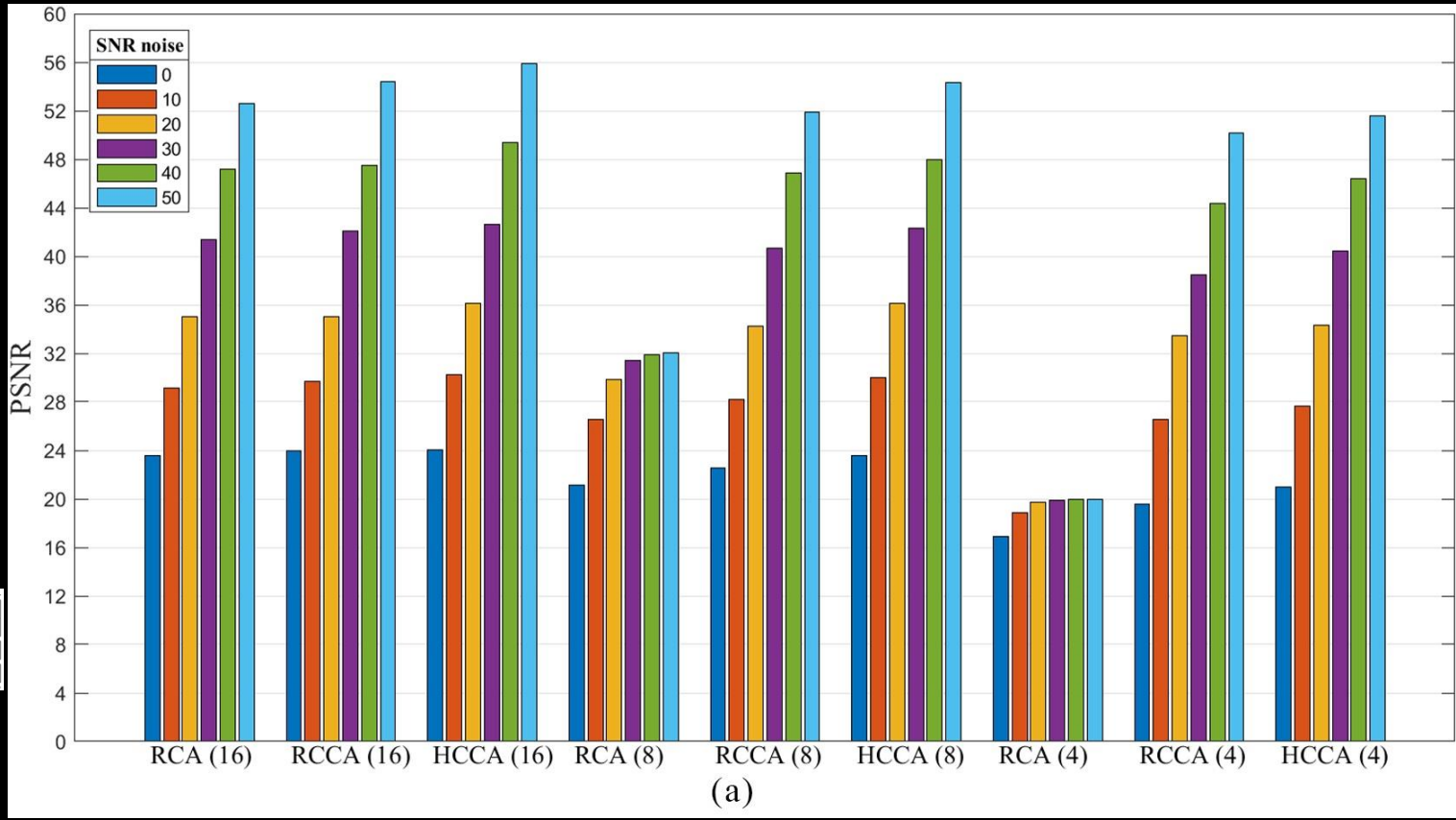
Simulation: Phase Quality (PSNR)

Parameters

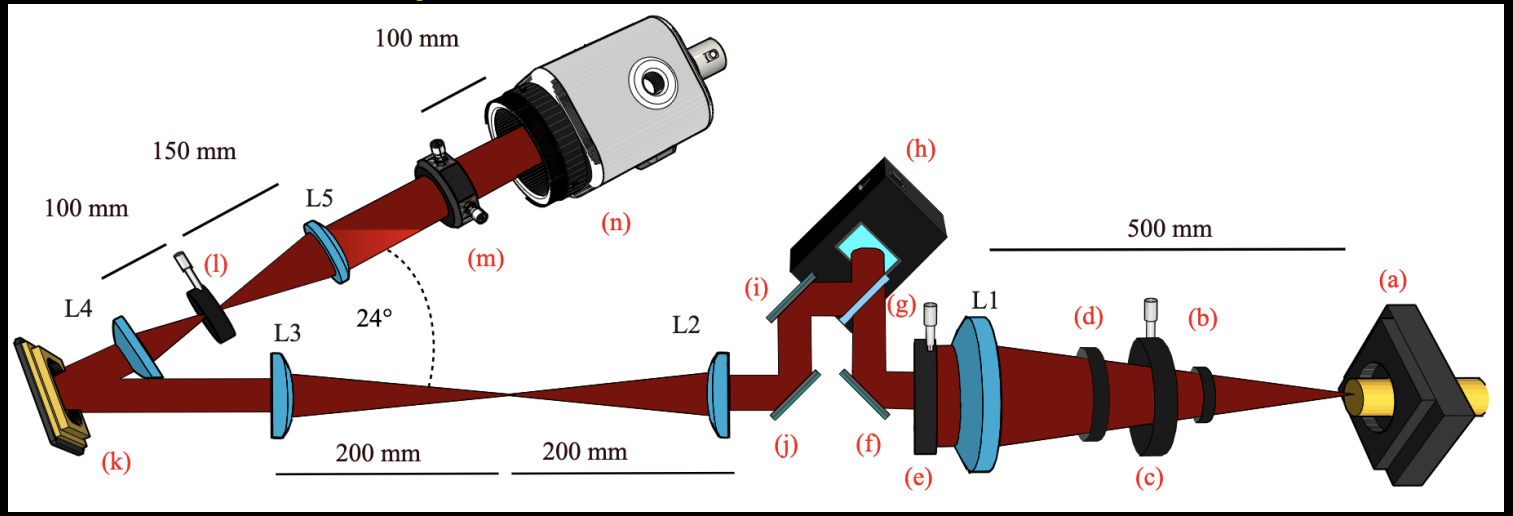
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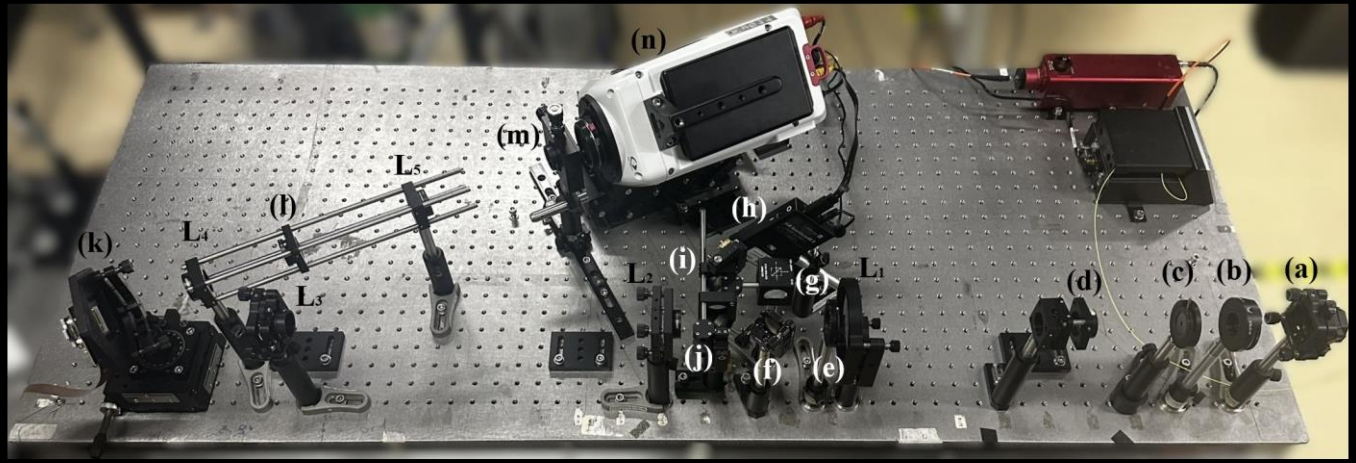


Experimental implementation



- (a) of λ = 642 nm propagating through different optical components as a linear polarizer (b);
- a half-waveplate (d); a collimator lens (L1); field stops (c), (e) and (l); mirrors (f), (i) and (j); a beam splitter (g); an SLM-LCoS (h);
- a 4f -system composed of lenses (L2) and (L3); an SLM-DMD (k); a 1.5× telescope system composed of lenses (L4) and (L5);
- a testing object (m) placed in a plane conjugated with the LCoS and DMD devices; and finally through a CMOS camera (n)

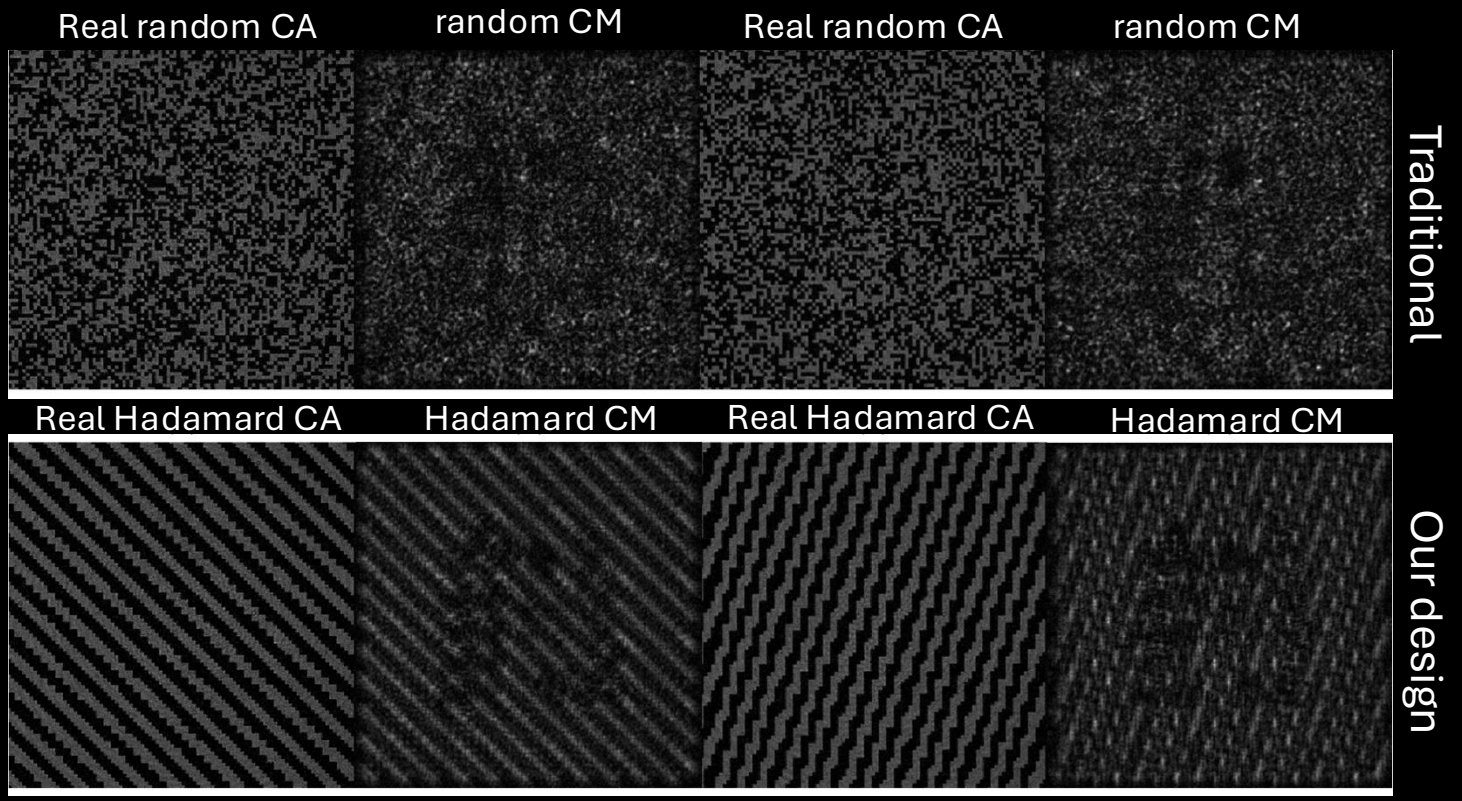
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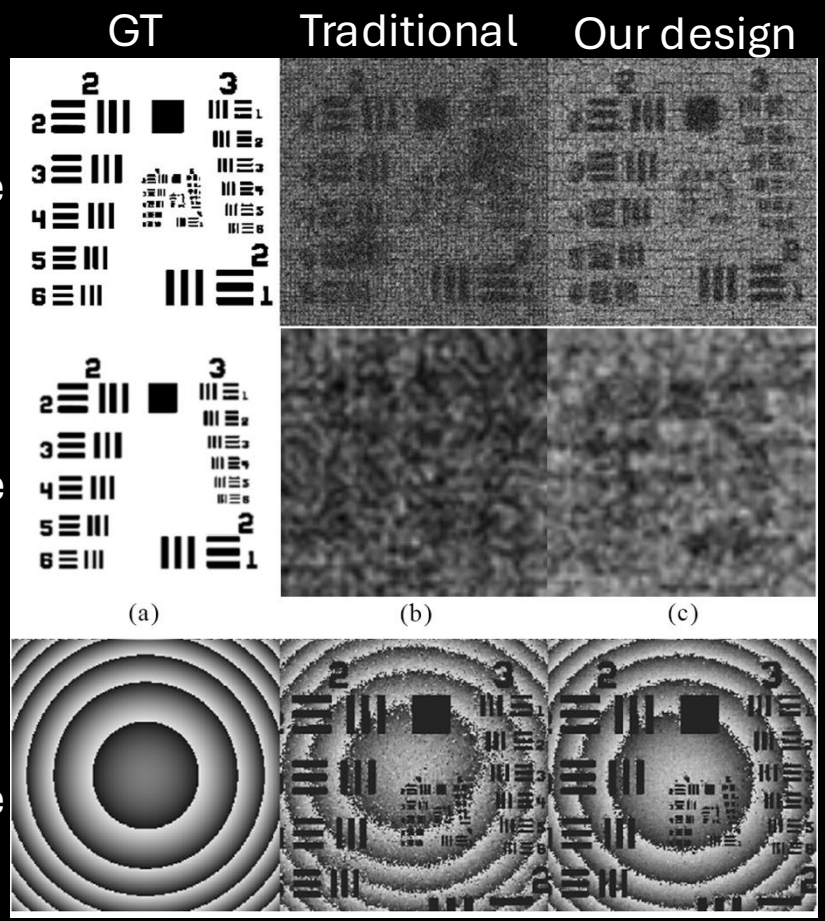
Propagation of Coded Apertures

CM: compressive measurement



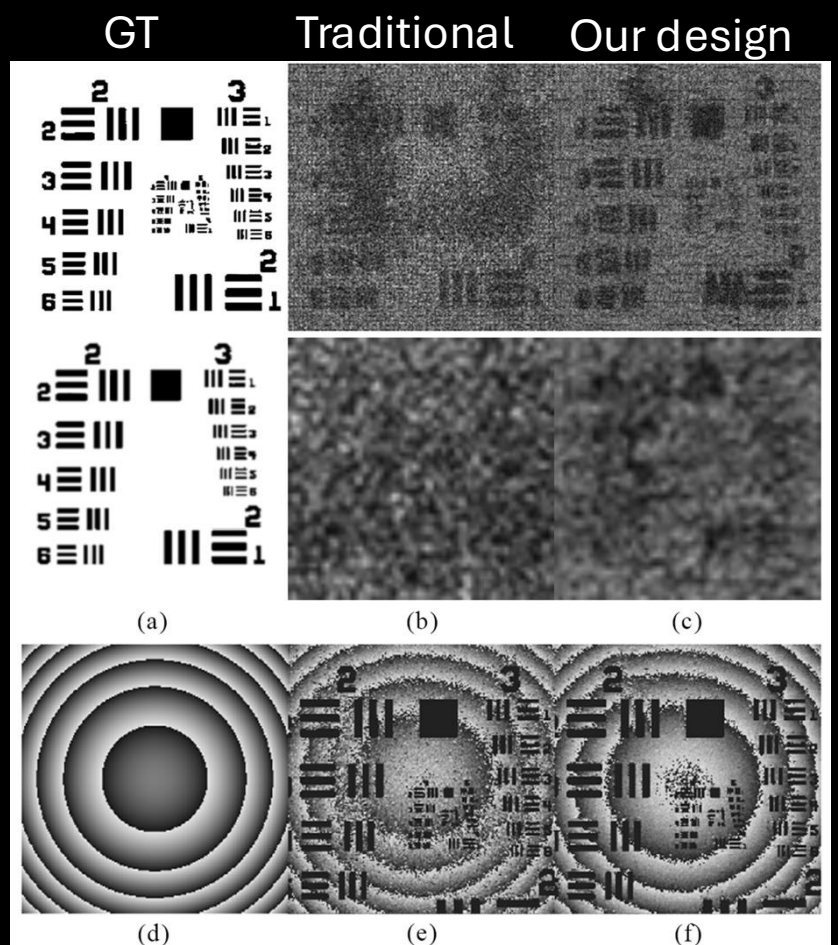
Real-data: Reconstructions

- Coded apertures: 16 RCCA, 16 HCCA
- Iterations: 100
- Reconstruction algorithm: HIO
- Amplitude: USAF 1951 benchmark
- Phase: 1000 mm lens

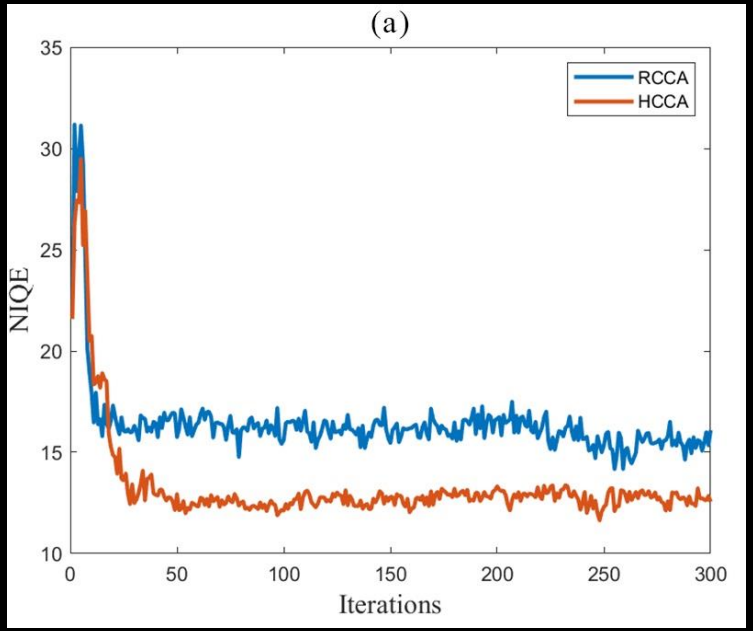


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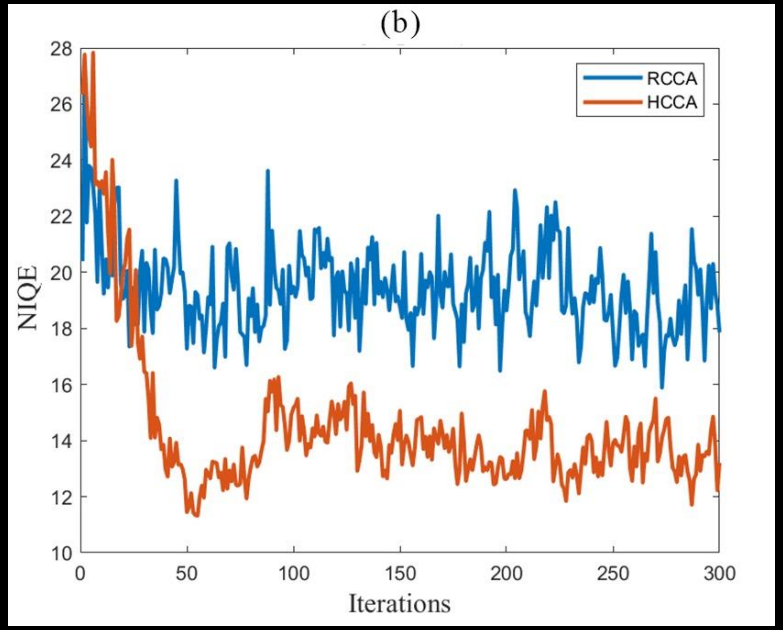
- Coded apertures: 8 RCCA, 8 HCCA
- Iterations: 1500
- Algorithm: HIO
- Reconstruction algorithm: Hybrid input-output
- Amplitude: USAF 1951 benchmark
- Phase: 1000 mm lens



Real-data: Phase and Amplitude Reconstruction



(a) Amplitude reconstruction with 16 CAs;

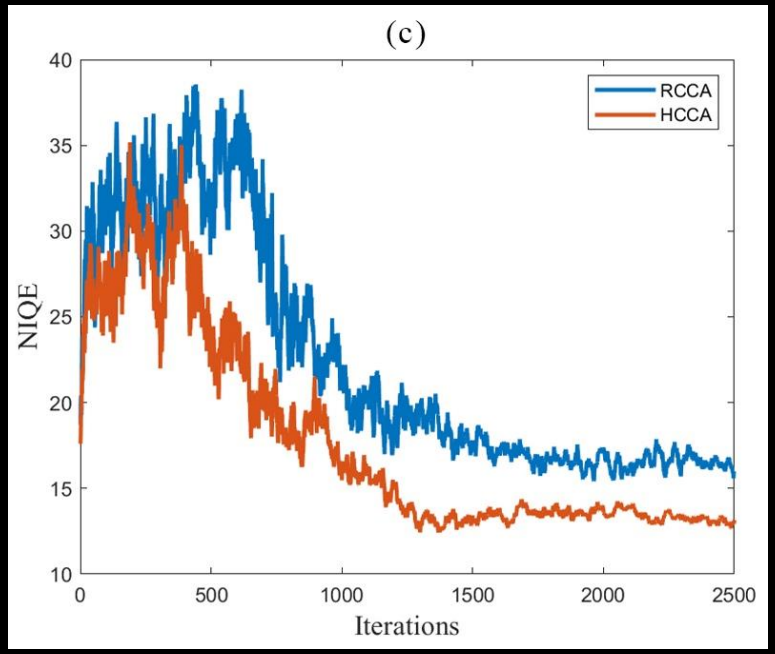


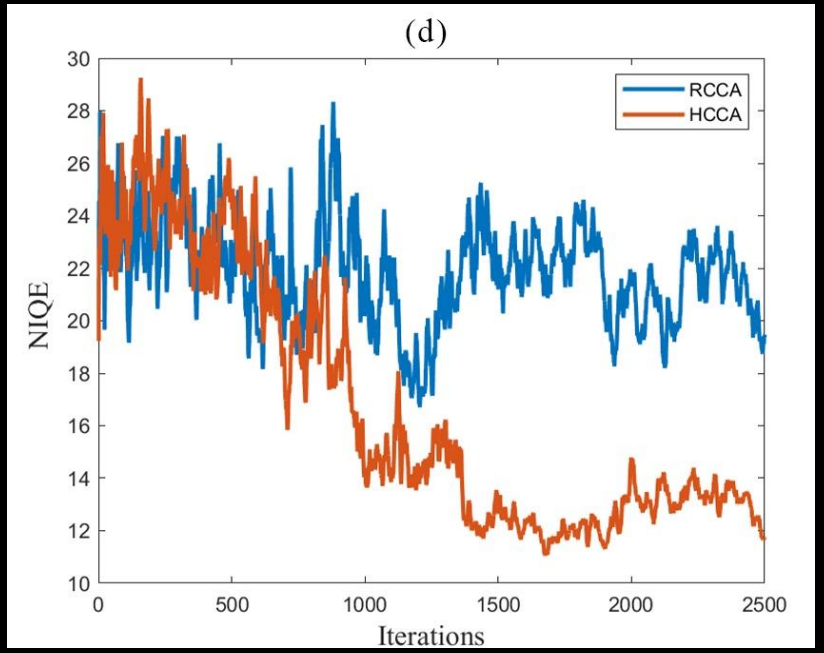
(b) phase reconstruction with 16 CAs;

Quality comparison via the NIQE between RCCA and HCCA for 16 apertures.

NIQE: naturalness image quality evaluator metric

Real-data: Phase and Amplitude Reconstruction





(c) amplitude reconstruction with 8 CAs;

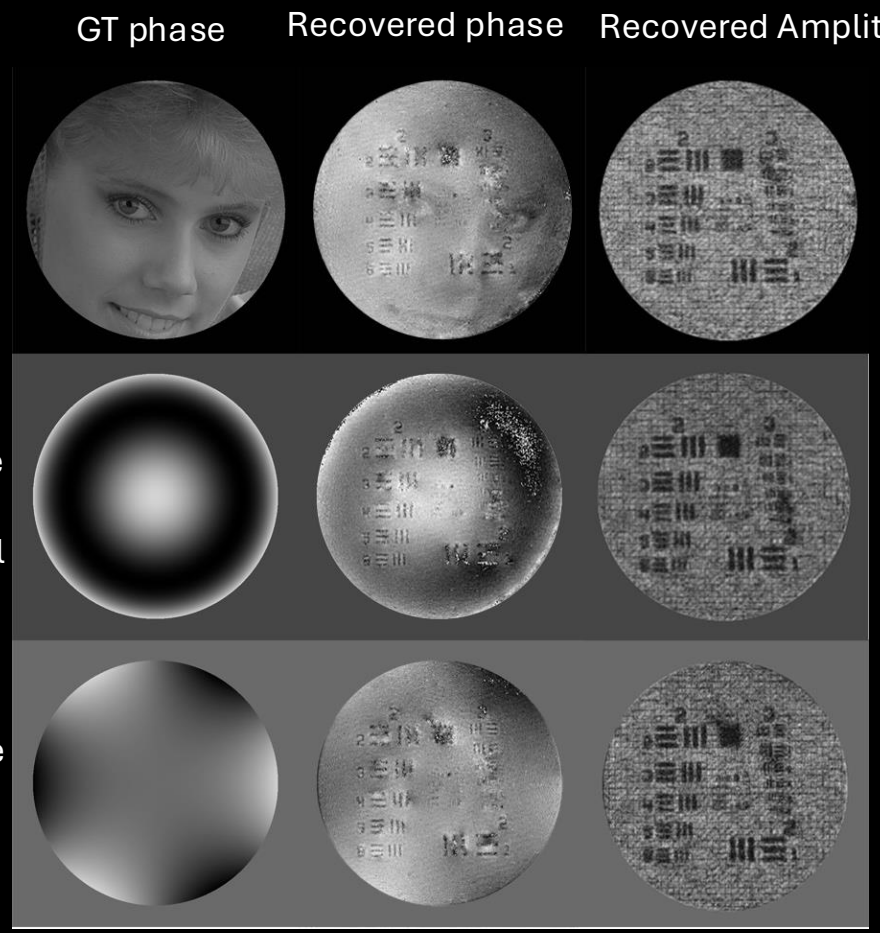
(d) phase reconstruction with 8 CAs.

Quality comparison via the NIQE between RCCA and HCCA for 8 apertures.

NIQE: naturalness image quality evaluator metric

Real-data: Phase retrieval

- Coded apertures: HCCA
- Algorithm: HIO
- Amplitude: USAF benchmark
- Phase: Kodak image, and two Zernikes

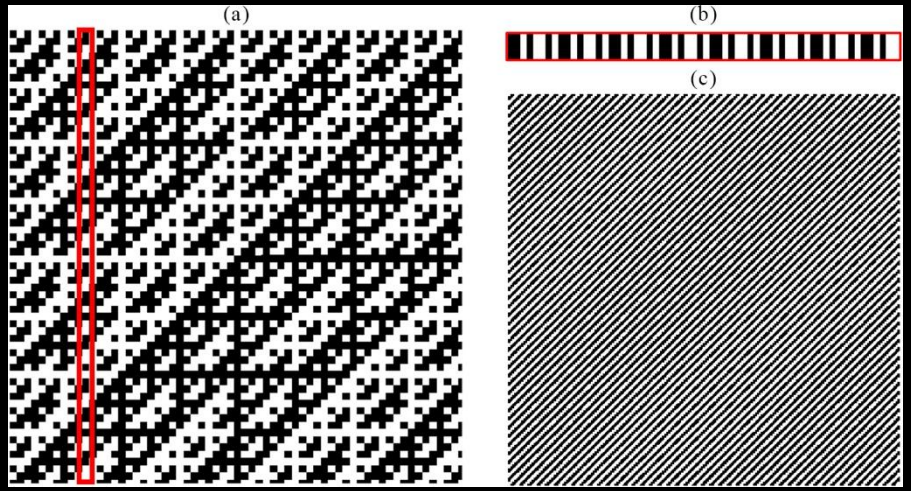


Conclusions

- We introduced a novel Hadamard complementary coded apertures for recovering the phase and amplitude of an unknown transmision target by encoding coherent light passing through it and measuring the intensity.

Three Advantages:

- A reduction in the number of coding masks
- Robustness to noise
- The ability to recovery arbitrary phases.
- Future work: Further reduction of mask and reconstruction algorithm



Hadamard complementary coded apertures

